

Mid Term Test
IT509 Data Mining and Data Warehousing
4 October, 2016
Marks: 25 Time: 1 hour

1. State true or false: 1×5=5
- a) The *SSE* (sum of squared error) measure of a *k*-means clustering result increases if number of clusters (*k*) is increased.
 - b) Single Linkage hierarchical clustering algorithm is more affected by noise.
 - c) Separating the data warehouse from the online database promotes high performance for both the systems.
 - d) Data stored in OLAP servers are summarized, consolidated.
 - e) A data warehouse performs day-to-day operational processing.
2. a) What do you understand by clustering in data mining? 2
- b) Mention some characteristics that a good clustering algorithm should possess. 2
- c) State important features for each of the following category of clustering algorithms: 2×3=6
Partitional algorithms, Hierarchical algorithms, Density-based algorithms.
3. a) Define the concept of density for the purpose of density-based clustering? 2
- b) Define a density based cluster utilizing your definition of density. 3
- c) What is a border object? Show a border object diagrammatically. 2
4. Briefly describe how to perform evaluation of the result produced by a clustering algorithm. 4
5. Perform the following clustering operations on the data set shown in the figure below:
- | | | | | |
|---|---|---|---|---|
| A | B | C | D | |
| • | • | • | • | |
| | | | | • |
| | | | | E |
| • | • | • | • | |
| F | G | H | I | |
- a) Show the dendrogram for Single Linkage hierarchical clustering. 3
- b) Show the dendrogram for Complete Linkage hierarchical clustering. 3
6. What is a data warehouse? Explain. 5
7. Describe the 3-tier architecture of a data warehouse. 3